Ticket Command Center: Customer Support Performance Report

Python and Power BI capstone. Public Kaggle dataset of 8,469 support tickets (Customer Support Tickets Dataset). Prepared by Mohamed Nashat.

Two in three tickets are not closed, only 39% of closed tickets are rated satisfied, and about half of the resolution timestamps are invalid. Ticket outcomes look the same across every priority, channel, type and product, so the issue is operational health and data capture, not a specific segment.

8,469	32.7%	2.99 / 5	49.3%
Tickets	Closed	Average CSAT	Closed tickets with invalid timestamps

Executive summary

Key findings

- **A large backlog.** 5,700 tickets (67.3%) are not closed: 2,881 are pending a customer response and 2,819 are open. Only 2,769 are closed.
- **Satisfaction is low.** The average rating is 2.99 out of 5, and 39.3% of rated tickets (1,087 of 2,769) are satisfied (4 or 5). Ratings are spread almost evenly from 1 to 5.
- **Outcomes are flat across segments.** The closed rate stays between 30% and 34% for every priority, channel, ticket type and gender. Satisfaction stays between 37% and 43%. No segment stands out.
- **Resolution timestamps cannot be trusted.** In 1,365 of 2,769 closed tickets (49.3%), the resolution is dated before the first response. Only 1,404 tickets have a usable resolution time (average 7.6 hours).

Recommendations

1. Clear the backlog: set reminders and a follow-up rule for the 2,881 tickets waiting on customers.
2. Fix timestamp capture: record ticket-created, first-response and resolved times, and reject a resolution earlier than the first response.
3. Reset SLA targets once timestamps are reliable. The 24-hour target cannot be violated in this data (see section 1).
4. Do not target a specific segment based on this data. Differences between segments are small.

1. Overview and SLA

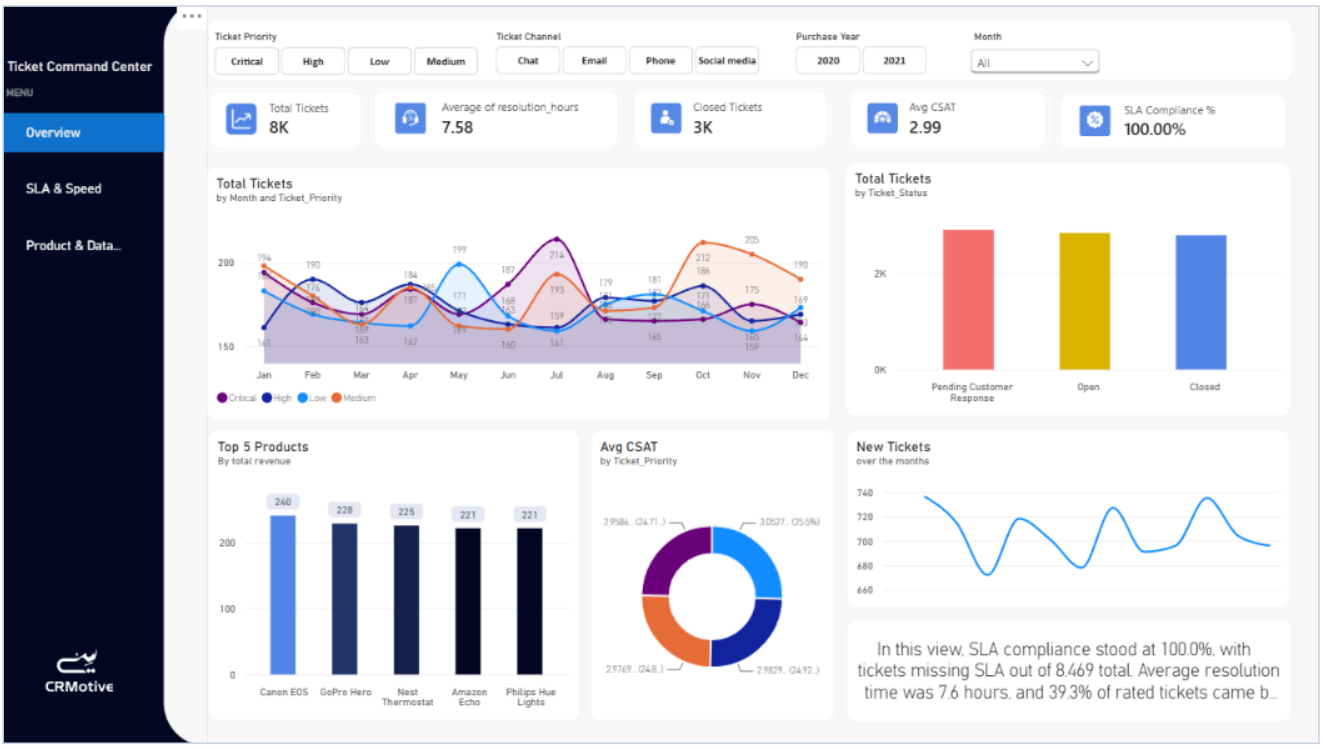


Figure 1. Overview page of the Power BI report. The Top 5 Products chart counts tickets, not revenue.

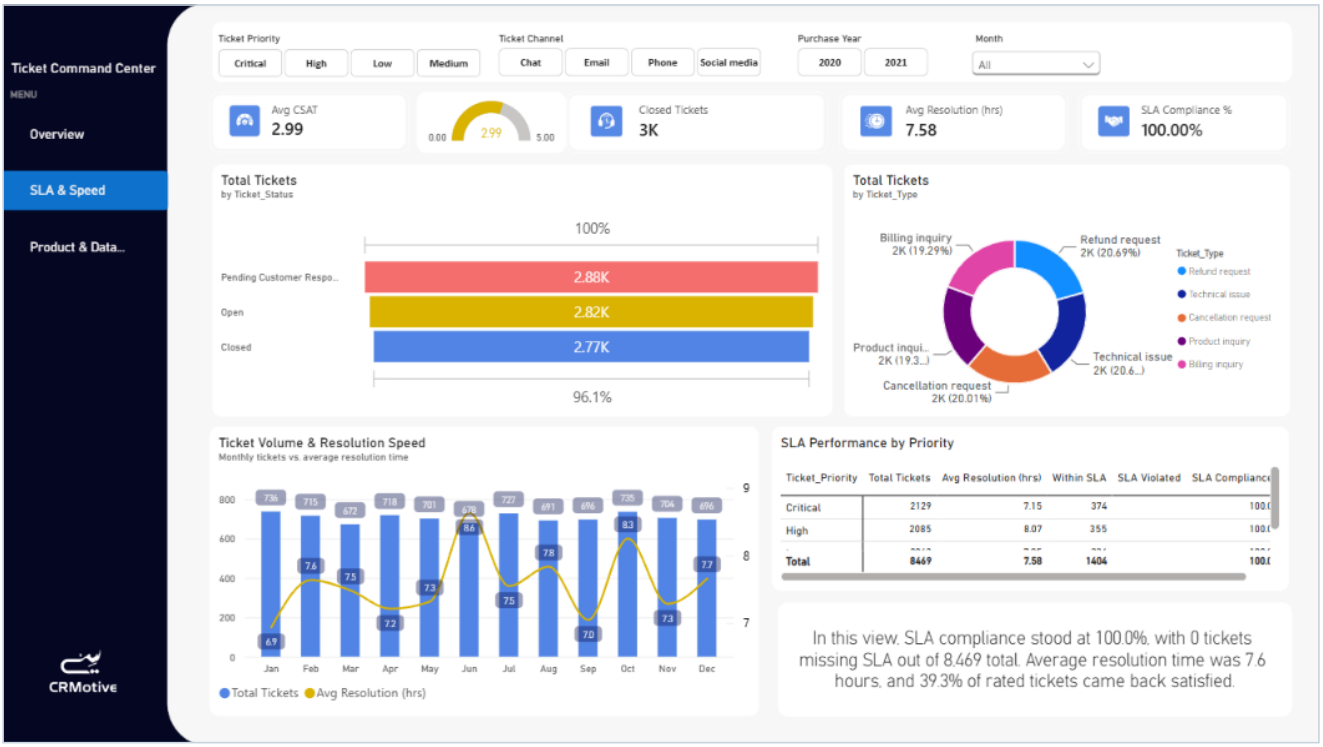


Figure 2. SLA and Speed page. SLA compliance reads 100% because the longest valid resolution (23.5 hours) is under the 24-hour target. It covers only the 1,404 tickets with valid timestamps.

- **Monthly volume is steady:** between 672 and 736 tickets a month, with no trend.
- **The three statuses are almost equal:** Pending Customer Response 2,881, Open 2,819, Closed 2,769.
- **Ticket types are balanced too:** each of the five types is about 20% of tickets.

2. Products and data quality

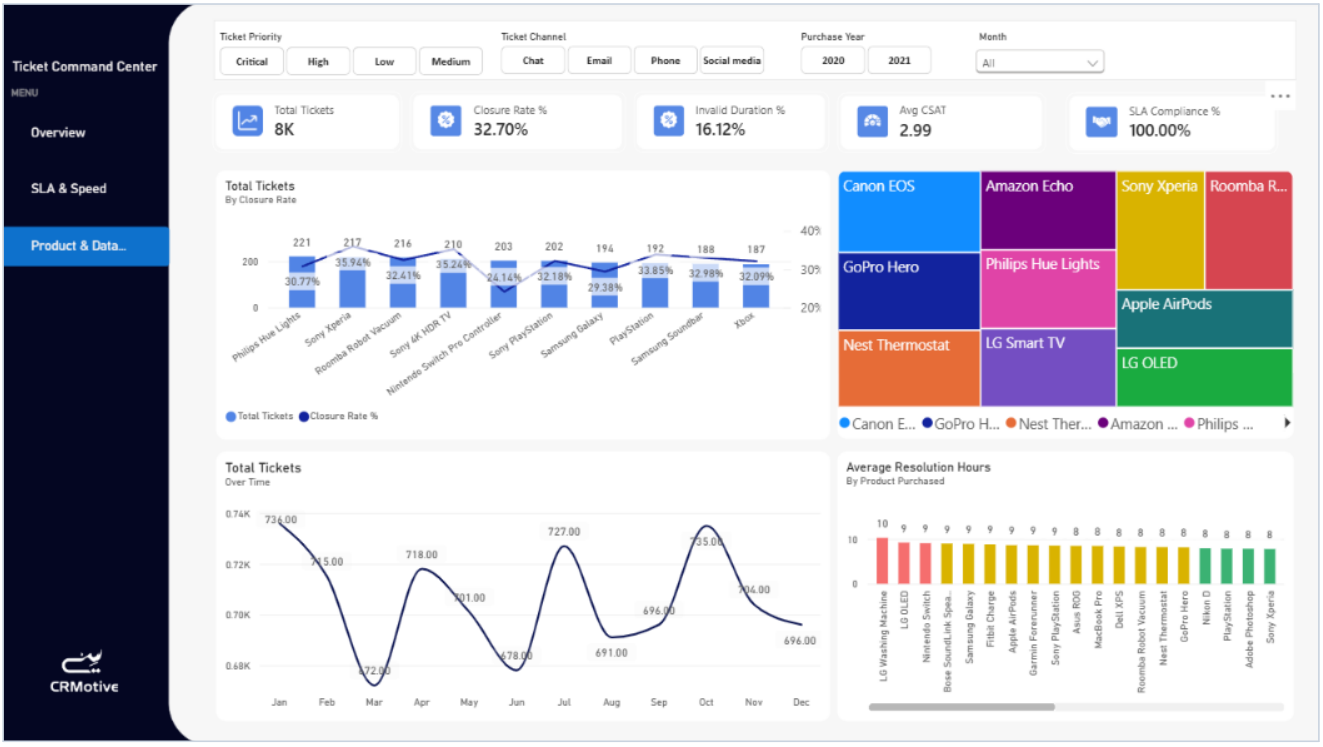


Figure 3. Product and Data Quality page. Invalid Duration % (16.1%) is measured on all tickets. On closed tickets only, the share is 49.3%.

- **Products behave alike.** The most-ticketed products (Canon EOS 240, GoPro Hero 228, Nest Thermostat 225, Amazon Echo 221, Philips Hue Lights 221) each have about 190 to 240 tickets, and closure rates between 24% and 36%. With samples this small, these gaps are not strong evidence.
- **Resolution time barely varies by product:** about 8 to 10 hours on average.
- **Faster resolution does not mean happier customers.** Satisfaction is 40.4% for the fastest quarter of tickets and 39.0% for the slowest.

3. Data quality, limitations and next steps

Cleaning decisions

Issue found	What was done
Resolution, resolution time and rating empty for 5,700 tickets	Left empty. They are missing only for tickets that are not closed, so filling them would invent results. They were not used to explain the Closed outcome (leakage).
1,365 closed tickets resolved before the first response	Kept in the data, flagged with invalid_duration, and excluded from resolution-time and SLA calculations.
No ticket-created timestamp	The first-response SLA cannot be measured. Monthly charts use the product purchase date.
Template tag left in descriptions	Replaced with the real product name.
Names and emails	Left out of the exported file.

Limitations

- **The data looks synthetic.** Emails use example.com, resolutions are random words, and every category has almost the same outcome. Treat the results as a demonstration of method, not as real customer behavior.
- **SLA compliance is not informative.** No valid ticket exceeds the 24-hour target, so the result is always 100%.
- **No agent, team or ticket-created time.** Workload, first-response speed and queue age cannot be measured.
- **Satisfaction exists only for closed tickets,** so it says nothing about the 67% still open.

Data to request

- Ticket created, first response and resolved timestamps, with validation.
- Assigned agent or team, so workload and performance can be compared.
- Customer feedback text, to explain the low satisfaction.